

Certificate of Analysis

Sample Name: LIT CULTURE BLACK HONEY
Client: Free River Distributing
Sample Code: DTS-260804-082
Matrix Name: Beverage & Liquid - Shot
Type / Result: Batch Release - Pass



Received Date: Thu, Aug 6, 2026
Published Date: Thu, Aug 13, 2026
Batch/Lot Code: BLKHNY-016
Batch Size: N/A
Sample Size: 1U
Average Unit Weight: 15.966g (Density (g/mL) * 15mL package. 3 servings/package)

RESULT SUMMARY

Mitragynine	28.24 mg /serv
Total Major Alkaloids	31.95 mg /serv

ALKU
Kratom Alkaloids
High Level

ALKL
Kratom Alkaloids
Low Level

SAL
Salmonella spp.
qPCR

ECOLI
Total Coliforms & E. coli
Plate

TAMC
Total Aerobic Bacteria
Plate

DEN
Density of Liquids

HVMET
Heavy Metals
Big 4

TYMFD
Total Yeast & Mold
Plate

SOLHM
Residual Solvents
National Panel

7OHFL
7-Hydroxymitragynine
Florida

Approvals

RESULTS REVIEWED BY:

Leslie Nagy
Laboratory Director

Cambium Analytica
Thursday, Aug 13, 2026

RESULTS CERTIFIED BY:

Douglas Smith
VP - Scientific Operations

Cambium Analytica
Thursday, Aug 13, 2026

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Lab Information

Address: 1230 Woodmere Ave, Traverse City, MI 49686
Phone: 231.252.3669
Accreditation: ISO/IEC 17025:2017 – #108157



ALKU Kratom Alkaloids - High Level LAB-TM-052 - Determination of Kratom Alkaloid Content by UPLC-DAD ALKU-DTS-260804-082-01 - MON, AUG 10, 2026 								
Analyte	Value	Value (mg/g)	Per Serving	Per Package	Action Limit	LOD	LOQ	Status
Mitragynine	0.5307 %	5.3068 mg/g	28.24 mg	84.73 mg	N/A	0.0517 mg/g	0.1034 mg/g	N / A
Paynantheine	0.0259 %	0.2593 mg/g	1.38 mg	4.14 mg	N/A	0.0517 mg/g	0.1034 mg/g	N / A
Speciogynine	0.0229 %	0.2286 mg/g	1.22 mg	3.65 mg	N/A	0.0517 mg/g	0.1034 mg/g	N / A
Speciociliatine	0.0209 %	0.2089 mg/g	1.11 mg	3.34 mg	N/A	0.0517 mg/g	0.1034 mg/g	N / A
Total Major Alkaloids *	0.6004 %	6.0036 mg/g	31.95 mg	95.85 mg	N/A	N/A	N/A	N / A

*Total Major Alkaloids is calculated as the sum of Mitragynine, Paynantheine, Speciociliatine and Speciogynine.

ALKL Kratom Alkaloids - Low Level LAB-TM-047 - Determination of Kratom Alkaloid Content by LC-TQ ALKL-DTS-260804-082-01 - MON, AUG 10, 2026 								
Analyte	Value	Value (mg/g)	Per Serving	Per Package	Action Limit	LOD	LOQ	Status
7-Hydroxymitragynine	0.00010 %	0.00104 mg/g	0.01 mg	0.02 mg	N/A	0.002 ug/g	0.011 ug/g	N / A
Mitraphylline	ND	N/A	N/A	N/A	N/A	0.004 ug/g	0.019 ug/g	N / A
Total Minor Alkaloids *	0.00010 %	0.00104 mg/g	0.01 mg	0.02 mg	N/A	N/A	N/A	N / A

*Total Minor Alkaloids is calculated as the sum of 7-Hydroxymitragynine and Mitraphylline.

SAL Salmonella spp. - qPCR - 25 g LAB-TM-063 - Qualitative Detection of Presumptive Salmonella spp. in Foods and Dietary Supplements SAL-DTS-260804-082-01 - MON, AUG 10, 2026 					
Analyte	Value	Action Limit	LOD	LOQ	Status
Salmonella spp.	ND /25 g	N/A	N/A	N/A	N / A

USP <62> - Vendor Modified, USP <2022> - Vendor Modified, USP <1223>, AOAC PTM 121802, AOAC OMA 2020.02

ECOLI Total Coliforms & E. coli - Plate - 25 g - Full Range LAB-TM-059 - Enumeration of Escherichia coli and Total Coliform in Foods and Dietary Supplements ECOLI-DTS-260804-082-01 - TUE, AUG 11, 2026 					
Analyte	Value	Action Limit	LOD	LOQ	Status
E. coli	ND	N/A	10 CFU/g	10 CFU/g	N / A
Total Coliforms	ND	N/A	10 CFU/g	10 CFU/g	N / A

USP <61> - Vendor Modified, USP <2021> - Vendor Modified, USP <1223>, AOAC PTM 051801, AOAC OMA 2018.13

Testing was conducted on a 25.6825 gram aliquot



TAMC Total Aerobic Bacteria - Plate - 25 g - Full Range LAB-TM-060 - Enumeration of Total Aerobic Count in Foods and Dietary Supplements TAMC-DTS-260804-082-01 - MON, AUG 10, 2026 					
Analyte	Value	Action Limit	LOD	LOQ	Status
Total Aerobic Count	ND	N/A	10 CFU/g	10 CFU/g	N / A

USP <61> - Vendor Modified, USP <1223>, AOAC PTM 121403, AOAC OMA 2015.13

Testing was conducted on a 25.6825 gram aliquot

DEN Density of Liquids LAB-TM-017 - Brix & Density Analysis DEN-DTS-260804-082-01 - FRI, AUG 7, 2026 					
Analyte	Value	Action Limit	LOD	LOQ	Status
Density	1.0644 g/mL	N/A	N/A	N/A	N / A
Specific Gravity *	1.0663	N/A	N/A	N/A	N / A

*Specific gravity is calculated using the density of water at 20 °C (0.9982 g/mL) using the equation: [Specific Gravity = (Density of sample in g/mL) ÷ 0.9982 g/mL]

HVMET Heavy Metals - Big 4 LAB-TM-044 - Determination of Heavy Metals by ICP-MS HVMET-DTS-260804-082-01 - MON, AUG 10, 2026 								
Analyte	Value	Value (mg/g)	Per Serving	Per Package	Action Limit	LOD	LOQ	Status
Arsenic	ND	N/A	N/A	N/A	N/A	9.341 ug/kg	25.947 ug/kg	N / A
Cadmium	ND	N/A	N/A	N/A	N/A	1.868 ug/kg	6.487 ug/kg	N / A
Lead	ND	N/A	N/A	N/A	N/A	1.142 ug/kg	6.487 ug/kg	N / A
Mercury	ND	N/A	N/A	N/A	N/A	0.363 ug/kg	0.649 ug/kg	N / A

USP <233>, USP <730>, USP <1730>

TYMFD Total Yeast & Mold - Plate - 25 g - Full Range LAB-TM-061 - Enumeration of Yeast and Mold in Foods and Dietary Supplements TYMFD-DTS-260804-082-02 - THU, AUG 13, 2026 					
Analyte	Value	Action Limit	LOD	LOQ	Status
Total Mold	ND	N/A	10 CFU/g	10 CFU/g	N / A
Total Yeast	11000 CFU/g	N/A	1000 CFU/g	1000 CFU/g	N / A
Total Yeast and Mold *	11000 CFU/g	N/A	N/A	N/A	N / A

*Total Yeast and Mold is calculated as the sum of Total Yeast and Total Mold

USP <61> - Vendor Modified, USP <1223>, AOAC PTM 121301, AOAC OMA 2014.05

Testing was conducted on a 1 gram aliquot





Analyte	Value	Action Limit	LOD	LOQ	Status
1,1-Dichloroethene	ND	N/A	5 ug/g	10 ug/g	N / A
1,2-Dichloroethane	ND	N/A	0.5 ug/g	1.00 ug/g	N / A
2-Methylbutane	ND	N/A	5 ug/g	10 ug/g	N / A
2-Methylpentane	ND	N/A	5 ug/g	10 ug/g	N / A
2,2-Dimethylbutane	ND	N/A	5 ug/g	10 ug/g	N / A
2,3-Dimethylbutane	ND	N/A	5 ug/g	10 ug/g	N / A
3-Methylpentane	ND	N/A	5 ug/g	10 ug/g	N / A
Acetone	ND	N/A	5 ug/g	10 ug/g	N / A
Acetonitrile	ND	N/A	5 ug/g	10 ug/g	N / A
Benzene	ND	N/A	0.18 ug/g	0.50 ug/g	N / A
Butane	ND	N/A	5 ug/g	10 ug/g	N / A
Chloroform	ND	N/A	0.78 ug/g	1 ug/g	N / A
Ethanol	2884.100 ug/g	N/A	5 ug/g	10 ug/g	N / A
Ethyl Acetate	33.274 ug/g	N/A	5 ug/g	10 ug/g	N / A
Ethyl Ether	ND	N/A	5 ug/g	10 ug/g	N / A
Ethylene Oxide	ND	N/A	2 ug/g	4 ug/g	N / A
Heptane	ND	N/A	5 ug/g	10 ug/g	N / A
Hexane	ND	N/A	5 ug/g	10 ug/g	N / A
Isobutane	ND	N/A	5 ug/g	10 ug/g	N / A
Isopropyl Alcohol	ND	N/A	5 ug/g	10 ug/g	N / A
Methanol	ND	N/A	5 ug/g	10 ug/g	N / A
Methylene Chloride	ND	N/A	5 ug/g	10 ug/g	N / A
Neopentane	ND	N/A	5 ug/g	10 ug/g	N / A
Pentane	ND	N/A	5 ug/g	10 ug/g	N / A
Propane	ND	N/A	5 ug/g	10 ug/g	N / A
Toluene	ND	N/A	5 ug/g	10 ug/g	N / A
Total Xylenes	ND	N/A	5 ug/g	10 ug/g	N / A
Trichloroethylene	ND	N/A	0.5 ug/g	1 ug/g	N / A
Total Butanes *	0.000 ug/g	N/A	N/A	N/A	N / A
Total Hexanes *	0.000 ug/g	N/A	N/A	N/A	N / A
Total Pentanes *	0.000 ug/g	N/A	N/A	N/A	N / A

*Total Butanes is calculated as the sum of Butane and Isobutane
*Total Hexanes is calculated as the sum of Hexane, 2,2-Dimethylbutane, 2,3-Dimethylbutane, 2-Methylpentane, and 3-Methylpentane.
*Total Pentanes is calculated as the sum of Pentane, 2-Methylbutane, and Neopentane.

Modified USP <467>



7OHFL

7-Hydroxymitragynine - Florida - Dry-Weight Basis

7-Hydroxymitragynine - (Florida — Dry-Weight Basis)

7OHFL-DTS-260804-082-01 - MON, AUG 10, 2026

Analyte	Value	Action Limit	LOD	LOQ	Status
Moisture (%) by Input *	65.000 %	N/A	N/A	N/A	N/A
7-Hydroxymitragynine - As Received *	1.04 ppm	N/A	N/A	N/A	N/A
7-Hydroxymitragynine - Dry-Weight *	2.98 ppm	400 ppm	N/A	N/A	PASS

*The Moisture (%) by Input result may represent either a client-provided input or a value measured by Cambium. This value is incorporated into associated calculated analytes for reporting purposes.

*7-Hydroxymitragynine – As Received: The “As Received” value represents the concentration of 7-hydroxymitragynine, expressed in parts per million (ppm), as reported in the Kratom Alkaloids – Low Level test. This corresponds to the measured percentage of 7-hydroxymitragynine in the sample as received.

*7-Hydroxymitragynine (Dry-Weight Basis): The dry weight concentration of 7-hydroxymitragynine is calculated using the following formula: $\text{Dry Weight 7-Hydroxymitragynine (ppm)} = \frac{\text{7-Hydroxymitragynine (ppm)} \times (100 / (100 - \text{Moisture by Input (\%)))}$

